



AUTHENTICATED KEY EXCHANGE PROTOCOLS USING IMAGE STEGANOGRAPHY

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Abstract:

Objective- We tend to study the matter of key institution for secure many-to-many communications. The problem is inspired by the proliferation of huge-scale distributed file systems supporting parallel access to multiple storage devices.

Limitations of Existing System- Our review of the existing Kerberos-based protocol shows that it has a number of limitations: (i) a metadata server facilitating key exchange between the clients and the storage devices has heavy workload that restricts the scalability of the protocol; (ii) the protocol does not provide forward secrecy; (iii) the metadata server generates itself all the session keys that are used between the clients and storage devices, and this inherently leads to key escrow.

Design / Methodology/ Approach- Our work focuses on the current Internet standard for such file systems, i.e., parallel Network File System (pNFS), which makes use of Kerberos to establish parallel session keys between clients and storage devices. Kerberos is a widely deployed network authentication protocol currently being considered for standardization. In this paper, we propose a variety of authenticated key exchange protocol that is designed to address the above issues. We show that our protocols are capable of reducing up to approximately 54% of the workload of the metadata server and concurrently supporting forward secrecy and escrow-freeness.

Practical implications- This paper is very helpful for those who require their data to be searched efficiently.

Originality/Value- This will help in improving the efficiency of the searching process by maintaining reliability in data storage of client.

Key words- Parallel sessions, authenticated key exchange, network file systems, forward secrecy, key escrow

I. INTRODUCTION

In the world of secret-key cryptography, a secure communication between two parties A and B can be thought of sending messages in an encrypted form using a secret commonly known to only A and B. This secret is referred to as a key, or in this case a secret key. This simple technique of encryption using a secret key can be dated back to ancient times. In the old times, the two parties A and B agreed on a common key by meeting in person and coming up with a secret. However, in today's world of computing, secure communications are often needed between parties. There are immediate problems in conducting a secure communication. The problems are (i) How does A know that she/he is really talking to B? (ii) How does A establish a key with B without seeing B in person? By the introduction of third party, certificates, and public key cryptography, the above problems can be solved. First problem is usually referred to as the problem of authentication and the second as key exchange. It's often desirable that a solution should be a protocol achieving both authentication and key exchange at the same time. This protocol is called Authenticated Key Exchange (AKE) protocol.

In this work, we investigate the problem of secure many-to-many communications in large-scale network file systems (NFSs) that support parallel access to multiple storage devices. That is, we consider a communication model where there is a large number of clients accessing multiple remote and distributed storage devices in parallel. Particularly, we focus on how to exchange key materials and establish parallel secure sessions between the clients and the storage devices in the parallel Network File System (pNFS)-the current Internet standard—in an efficient and scalable manner.

Our primary goal in this work is to design efficient and secure authenticated key exchange protocols that meet specific requirements of pNFS. Particularly, we attempt to meet the following desirable properties, which either have not been satisfactorily achieved or not achievable by the current Kerberos-based solution.

- Scalability-The metadata server facilitating access requests from a client to multiple storage devices should bear as little workload as possible such that the server will not become a performance bottleneck, but is capable of supporting a very large number of clients.
- Forward secrecy-the protocol should guarantee the security of past session keys when the long term secret key of a client or a storage device is compromised.
- Escrow free-the metadata server should not learn any information about any session key used by the client and the storage device, provided there is no collusion among them.

This paper is organized as follows- Section II talks about the related works, section III elaborates the proposed system, section IV tells about modules used in our proposed method and the paper ends with the conclusion and references.

II. RELATED WORKS

A variety of methods exist for authenticated key exchange in parallel network file system. Numerous works have been done to improve the various aspects of key exchange.

Michal Abdalla, Fabrice Benhamouada, Philip Mackenzie (2015) proposed an efficient password-authenticated key exchange protocol that is included in the OpenSSL library and is currently being used in practice. This protocol is a well-known and accepted model for authenticated key exchange protocol that incorporates online and offline password guessing, concurrent sessions, forward secrecy, server compromise, and loss of session keys. It also satisfies the strong security assumption in a model with algebraic adversaries. The original J-PAKE paper claimed to give a proof of security, but as pointed out by Katz, the proof was not in one of the well-known accepted models for authenticated key exchange and simply proved some ad-hoc properties in an isolated setting. This paper presents a proof of security for the J-PAKE protocol in the well-known authenticated key exchange model. The first remark that our reduction for J-PAKE is algebraic: in each protocol of the proof are algebraic.

Weiqiang Wen, Libin Wang, Jiaxin Pan (2015) proposed most widely accepted models in the security proofs of Authenticated Key exchange protocols are the Canetti-Krawczyk and extended Canetti Krawczyk models that admit different adversarial queries with ambiguities and incomparable strength. The contribution of the paper is two-fold. Firstly, it propose a new

security model of AKE with powerful and specific adversarial capabilities, which is roughly a combination of the CK and ECK models and named as PACK, which stands for Powerful and specific adversarial capabilities CK model and also means that our model PACKs the CK and ECK models into a single model. Secondly, present a new AKE protocol called VOAKE which is variant of the OAKE protocol.

Jinguo Li, Mi Wen, Tao Zhang (2016) proposed a group based authentication and key agreement in MTC networks. Machine type communication (MTC) is an important mobile communication approach in the long-term evolution advanced (LTE-A) networks. To meet the MTC security requirements, the access authentication processing of MTC devices needs to follow the evolved packet system-authentication and key agreement (EPS-AKA). Compared with other authentication protocols in the LTE-A networks, this method could not only authenticate several MTC devices simultaneously but also dynamically update the access-policy to control the access authority of MTC devices. This scheme has efficiently reduced the network traffic, but it does not support access authentication. It first compares our GR-AKA with prior existing related work in terms of bandwidth overhead and computation overhead. GR-AKA provides a dynamic access policy updating mechanism by changing authentication tokens on each authorized MTCD with session keys. Live migration, dynamic job placement and load balancing are not addressed in the work.

Method proposed by Andrew Chi-Chi Yao and Yunlei Zhao. Key-exchange, in particular Diffie-Hellman key exchange (DHKE), is among the core cryptographic mechanisms for ensuring network security. For key-exchange over the Internet, both security and privacy are desired. In this paper, it develop a family of privacy-preserving authenticated DHKE protocols named deniable Internet key-exchange (DIKE), both in the traditional PKI setting and in the identity-based setting. The newly developed DIKE protocols are of conceptual clarity and practical efficiency. They provide useful privacy protection to both protocol and participants. The PFS property of our DIKE protocol is from the following observations: (1) the computation of the session-key HK does not involve player's secret-keys. Thus, if the attacker A does not know x or y , under the CDH assumption it cannot derive the session-key even if both the two players secret-keys are compromised. (2) If X is just generated by the attacker, it nevertheless cannot successfully finish the session with B by providing a correct authenticate value NMZK.

III. PROPOSED METHOD

The figure represents the system design of the proposed system. The main difference in this design with the existing system is steganography. Here we encrypt session key to image. The existing system has a metadata server facilitating key exchange between client and the storage devices that has heavy workload that restricts the scalability of protocol. This protocol does not provide forward secrecy. The metadata server generates itself all the session keys that are used between the client and storage devices, and this inherently leads to key escrow.

Our proposed system provides forward secrecy. The metadata server executing our protocol has much lower workload than that of Kerberos-based approach. We have designed a protocol which not only provides forward secrecy, but is also escrow free.

Upload the image and encrypt key to image and the download image to the user. The image is uploaded to the user by the system. The server generates a session key to the file system. Verifies the session key in the image which is uploaded by the user with the session key generated by the server to the file system. If verified file access is provided else access denied.

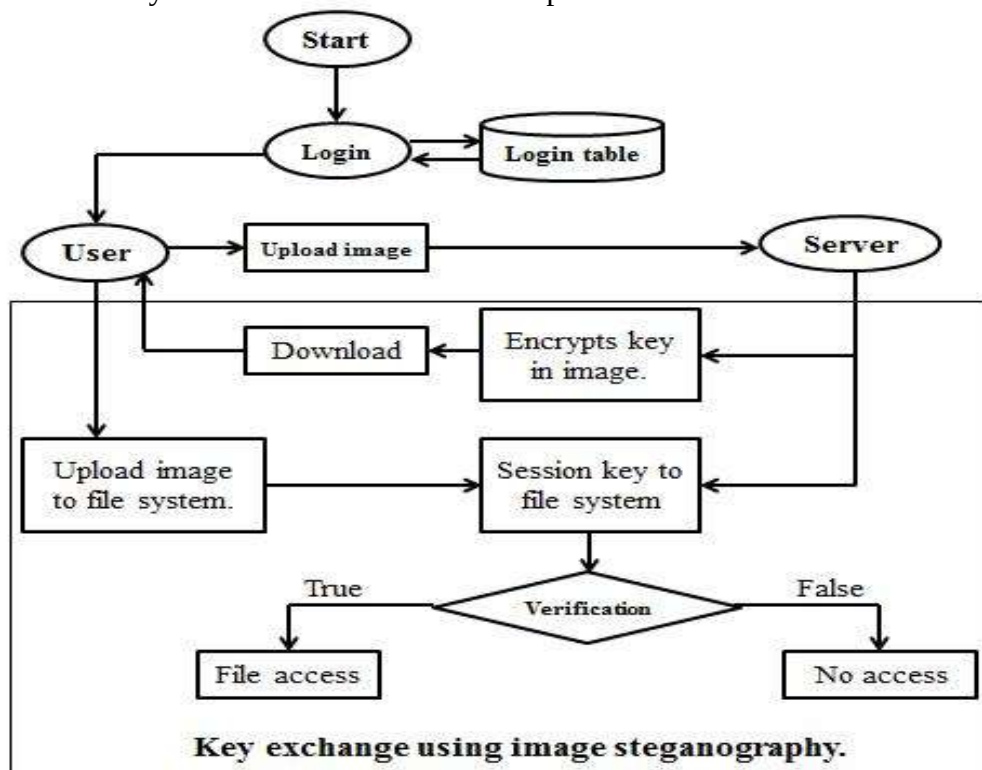


Figure: System architecture

IV. MODULE DESCRIPTION

Authenticated key exchange protocol for pNFS using image steganography technique consists of three modules.

- Parallel sessions.
- Authenticated key exchange.
- Validation.
- Escrow free-the metadata server should not learn any information about any session

4.1 PARALLEL SESSIONS

The parallel sessions are the parallel secure sessions between the clients and storage devices which are in parallel network file system (pNFS). Here the user logs in and the login details are obtained from the login table. The user then uploads image to the server. Then it will download the encrypted image from the server to the user.

4.2 AUTHENTICATED KEY EXCHANGE

The primary goal in this work is to design the efficient and secure authenticated key exchange protocol that meets the specific requirements of pNFS. Three new provably secure authenticated key exchange protocols are the main results of this paper. The server encrypts session key to the image by the technique of image steganography. The image is then downloaded by the user. The server generates session key to the file system for verification.

4.3 VALIDATION

The module includes validation process. Here the user uploads the encrypted image to the file system. The image is then verified with the session key generated by the server to the file system. If the image is verified access to file is permitted else access is denied.

V. CONCLUSION

We proposed three authenticated key exchange protocol for parallel network file system using image steganography. Our protocols offer three advantages over the existing Kerberos-based protocol system. Firstly, metadata server executing our protocol has much lower workload than that of Kerberos-based approach. Secondly, protocol provides forward secrecy. Thirdly, the protocol not only provides forward secrecy but also is escrow free.

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